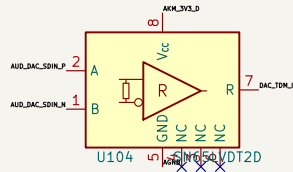
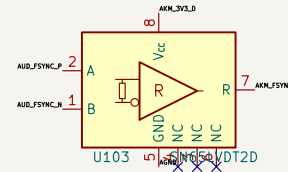
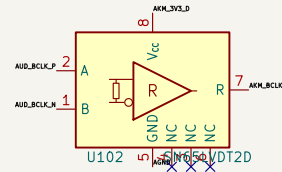
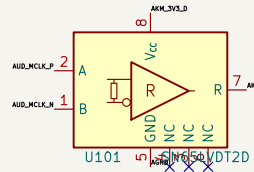
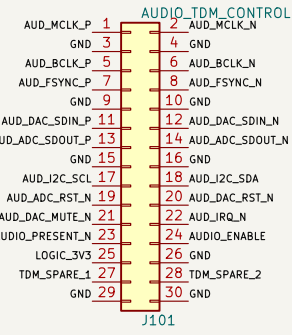




1. LOCKED CARRIER HARNESS

2. TERMINATED LVDS RECEIVERS



100 ohm termination

100 ohm termination

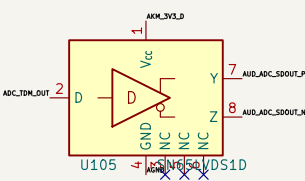
100 ohm termination

100 ohm termination

Pin lengths and pin assignment are controlled by dts/vhdl/lbcs/headers.at100

3. ADC RETURN LVDS DRIVER

4. CONTROL DEFAULTS AND POWER STATE



Reset and mode default asserted with the harness is absent or the carrier is equipped.

AD20L2EMBL is consumed by the local audio-power sequence; firmware releases asserts only after all PS signals and clocks are valid.

Route MCLK, M0M0S0, M0M0FSYN, DAC0M0N, and ADC0M0OUT as short referenced single-ended traces after the transistors.



## 1. AK5558VN EXACT PIN CAPTURE

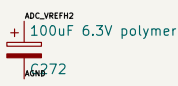
## 2. MODE STRAPS

|          |       |       |          |       |       |
|----------|-------|-------|----------|-------|-------|
| AF041616 | ADP1  | ADP1  | AF041616 | ADP1  | ADP1  |
| AF041617 | ADP2  | ADP2  | AF041617 | ADP2  | ADP2  |
| AF041618 | ADP3  | ADP3  | AF041618 | ADP3  | ADP3  |
| AF041619 | ADP4  | ADP4  | AF041619 | ADP4  | ADP4  |
| AF041620 | ADP5  | ADP5  | AF041620 | ADP5  | ADP5  |
| AF041621 | ADP6  | ADP6  | AF041621 | ADP6  | ADP6  |
| AF041622 | ADP7  | ADP7  | AF041622 | ADP7  | ADP7  |
| AF041623 | ADP8  | ADP8  | AF041623 | ADP8  | ADP8  |
| AF041624 | ADP9  | ADP9  | AF041624 | ADP9  | ADP9  |
| AF041625 | ADP10 | ADP10 | AF041625 | ADP10 | ADP10 |
| AF041626 | ADP11 | ADP11 | AF041626 | ADP11 | ADP11 |
| AF041627 | ADP12 | ADP12 | AF041627 | ADP12 | ADP12 |
| AF041628 | ADP13 | ADP13 | AF041628 | ADP13 | ADP13 |
| AF041629 | ADP14 | ADP14 | AF041629 | ADP14 | ADP14 |
| AF041630 | ADP15 | ADP15 | AF041630 | ADP15 | ADP15 |
| AF041631 | ADP16 | ADP16 | AF041631 | ADP16 | ADP16 |
| AF041632 | ADP17 | ADP17 | AF041632 | ADP17 | ADP17 |
| AF041633 | ADP18 | ADP18 | AF041633 | ADP18 | ADP18 |
| AF041634 | ADP19 | ADP19 | AF041634 | ADP19 | ADP19 |
| AF041635 | ADP20 | ADP20 | AF041635 | ADP20 | ADP20 |
| AF041636 | ADP21 | ADP21 | AF041636 | ADP21 | ADP21 |
| AF041637 | ADP22 | ADP22 | AF041637 | ADP22 | ADP22 |
| AF041638 | ADP23 | ADP23 | AF041638 | ADP23 | ADP23 |
| AF041639 | ADP24 | ADP24 | AF041639 | ADP24 | ADP24 |
| AF041640 | ADP25 | ADP25 | AF041640 | ADP25 | ADP25 |
| AF041641 | ADP26 | ADP26 | AF041641 | ADP26 | ADP26 |
| AF041642 | ADP27 | ADP27 | AF041642 | ADP27 | ADP27 |
| AF041643 | ADP28 | ADP28 | AF041643 | ADP28 | ADP28 |
| AF041644 | ADP29 | ADP29 | AF041644 | ADP29 | ADP29 |
| AF041645 | ADP30 | ADP30 | AF041645 | ADP30 | ADP30 |
| AF041646 | ADP31 | ADP31 | AF041646 | ADP31 | ADP31 |
| AF041647 | ADP32 | ADP32 | AF041647 | ADP32 | ADP32 |
| AF041648 | ADP33 | ADP33 | AF041648 | ADP33 | ADP33 |
| AF041649 | ADP34 | ADP34 | AF041649 | ADP34 | ADP34 |
| AF041650 | ADP35 | ADP35 | AF041650 | ADP35 | ADP35 |
| AF041651 | ADP36 | ADP36 | AF041651 | ADP36 | ADP36 |
| AF041652 | ADP37 | ADP37 | AF041652 | ADP37 | ADP37 |
| AF041653 | ADP38 | ADP38 | AF041653 | ADP38 | ADP38 |
| AF041654 | ADP39 | ADP39 | AF041654 | ADP39 | ADP39 |
| AF041655 | ADP40 | ADP40 | AF041655 | ADP40 | ADP40 |
| AF041656 | ADP41 | ADP41 | AF041656 | ADP41 | ADP41 |
| AF041657 | ADP42 | ADP42 | AF041657 | ADP42 | ADP42 |
| AF041658 | ADP43 | ADP43 | AF041658 | ADP43 | ADP43 |
| AF041659 | ADP44 | ADP44 | AF041659 | ADP44 | ADP44 |
| AF041660 | ADP45 | ADP45 | AF041660 | ADP45 | ADP45 |
| AF041661 | ADP46 | ADP46 | AF041661 | ADP46 | ADP46 |
| AF041662 | ADP47 | ADP47 | AF041662 | ADP47 | ADP47 |
| AF041663 | ADP48 | ADP48 | AF041663 | ADP48 | ADP48 |
| AF041664 | ADP49 | ADP49 | AF041664 | ADP49 | ADP49 |
| AF041665 | ADP50 | ADP50 | AF041665 | ADP50 | ADP50 |
| AF041666 | ADP51 | ADP51 | AF041666 | ADP51 | ADP51 |
| AF041667 | ADP52 | ADP52 | AF041667 | ADP52 | ADP52 |
| AF041668 | ADP53 | ADP53 | AF041668 | ADP53 | ADP53 |
| AF041669 | ADP54 | ADP54 | AF041669 | ADP54 | ADP54 |
| AF041670 | ADP55 | ADP55 | AF041670 | ADP55 | ADP55 |
| AF041671 | ADP56 | ADP56 | AF041671 | ADP56 | ADP56 |
| AF041672 | ADP57 | ADP57 | AF041672 | ADP57 | ADP57 |
| AF041673 | ADP58 | ADP58 | AF041673 | ADP58 | ADP58 |
| AF041674 | ADP59 | ADP59 | AF041674 | ADP59 | ADP59 |
| AF041675 | ADP60 | ADP60 | AF041675 | ADP60 | ADP60 |
| AF041676 | ADP61 | ADP61 | AF041676 | ADP61 | ADP61 |
| AF041677 | ADP62 | ADP62 | AF041677 | ADP62 | ADP62 |

AK5558VN  
U201

### 3. REFERENCES AND LOCAL BYPASS

Steps establish the safe hardware default: firmware still writes and verifies the serial-control registers after reset



QFN exposed-pad geometry and assembly coupon remain a routing release gate  
Each VREFH uses 20 ohm plus 100 nF and 100 uF to its VREFL/AGND  
AVDD=5.0 V, TVDD=3.3 V, internal 1.8 V LDO enabled  
TDM256, 32-bit I2S, slave mode: TDM1:0=10, DIF1:0=11, MSN=0

ProComm  
Sheet: /  
File: AK5558-ADC.kicad\_sch  
**Title: ProComm AUDIO-8X8 - AK5558VN Eight-Channel ADC**  
Size: A1      Date: 2026-08-16      Rev: A1



## 1. AK4458VN EXACT PIN CAPTURE

## 2. SERIAL-CONTROL DEFAULTS

[illegible]

CR01:B+01 selects DAC address 0x11; AOC remains at 0x10. TIM256 and 32-bit IQ5 are written only while route is asserted.

The unused S0TQ/3/4 and OSD pins are tied low in PCM TDM mode to prevent floating inputs.

### 3. REFERENCES AND LOCAL BYPASS



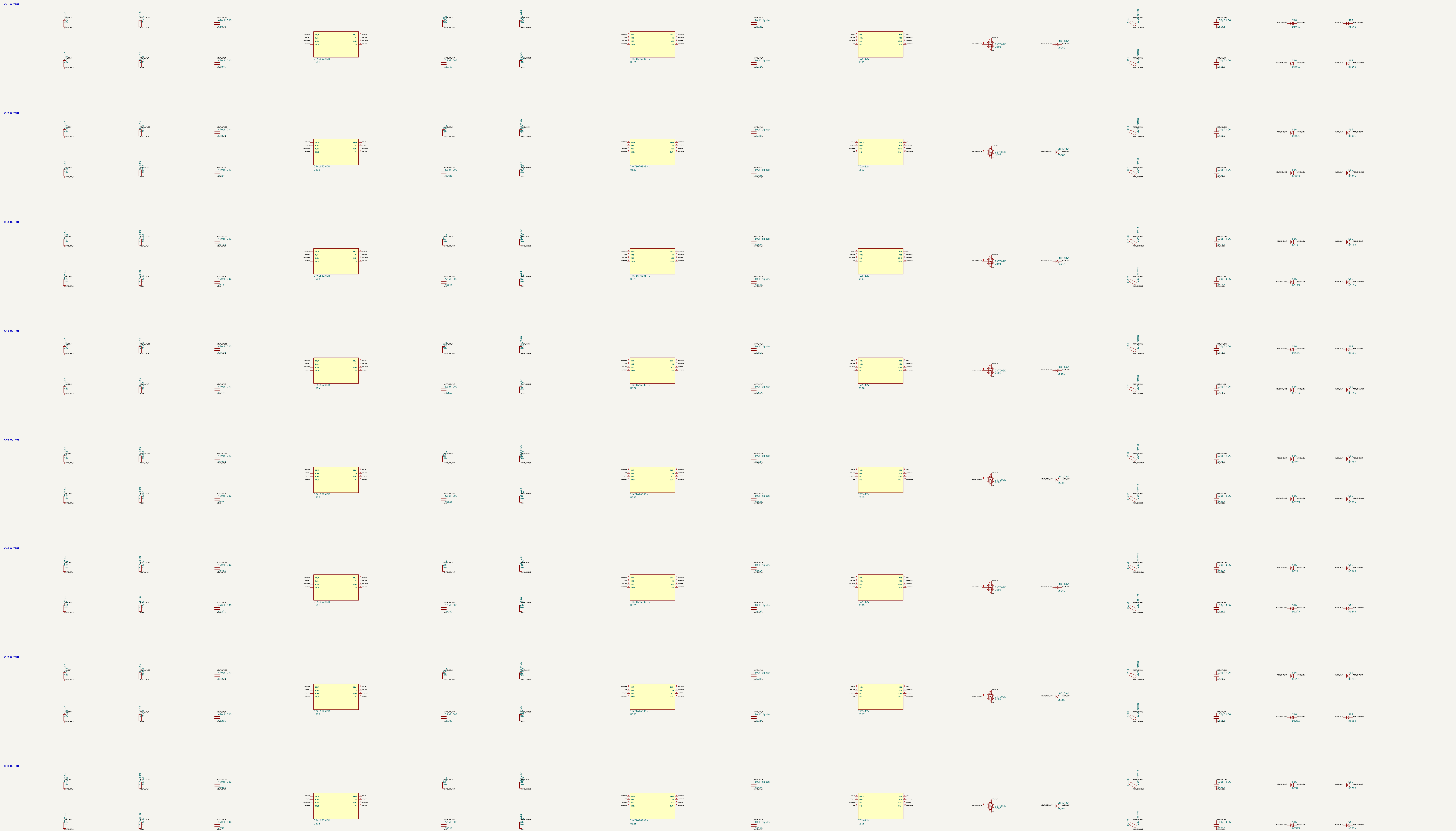




EIGHT IDENTICAL OUTPUT CHANNELS – SIGNAL FLOWS LEFT TO RIGHT

Delays may occur only after all audio PG signals, TDM clocks, resets, and DAC control sequencing pass.

OVER\_FUS



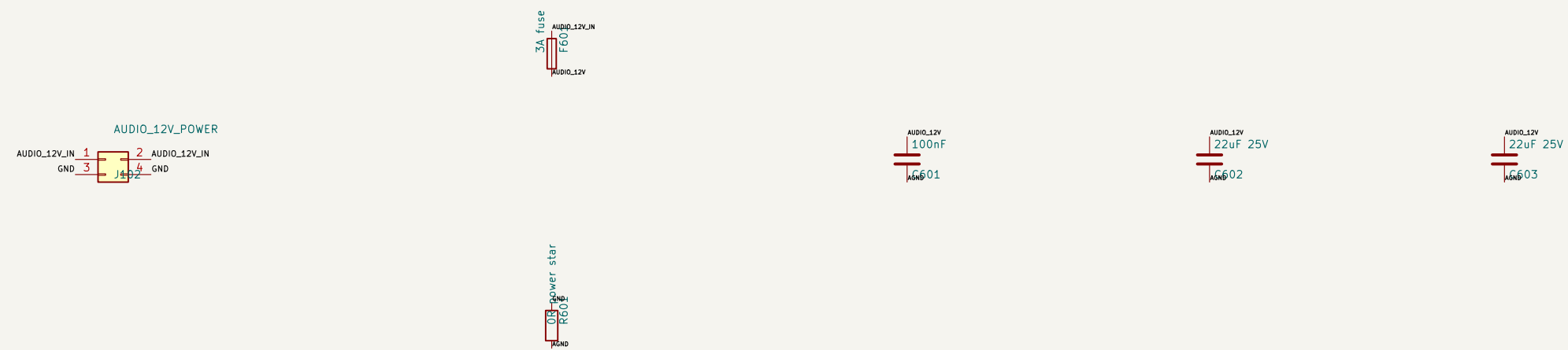
Output: REUSE: GPT5: 36.5MB/10% SHC, max sample/transfer coupon, +2% eff. 36.5 into 330 atm and 30 kbar, short/open recovery, 30 injection, 150/100, and power-transfer pay/click mode

+24 dBu into 600 ohm is a qualification target; simulation and bench release require TO2-12V DPDT relays short both XLR legs to AGND while de-energized OPA1652 gain stage and THA1164B active-balanced line driver

AK4588 datasheet PCM DFF: 3.9 k, 4.7 k, 150 k, 470 pF, 3.9 nF  
 ProCom  
 Sheet: /  
 File: Audio-Outputs.kicad\_sch  
**Title: ProCom AUDIO-8X8 - Eight Balanced Output Channels**  
 Size: A0 Date: 2026-08-16 Res: A1  
 KiCad 8.0.A.10.0.5-rc1 No: 1/1



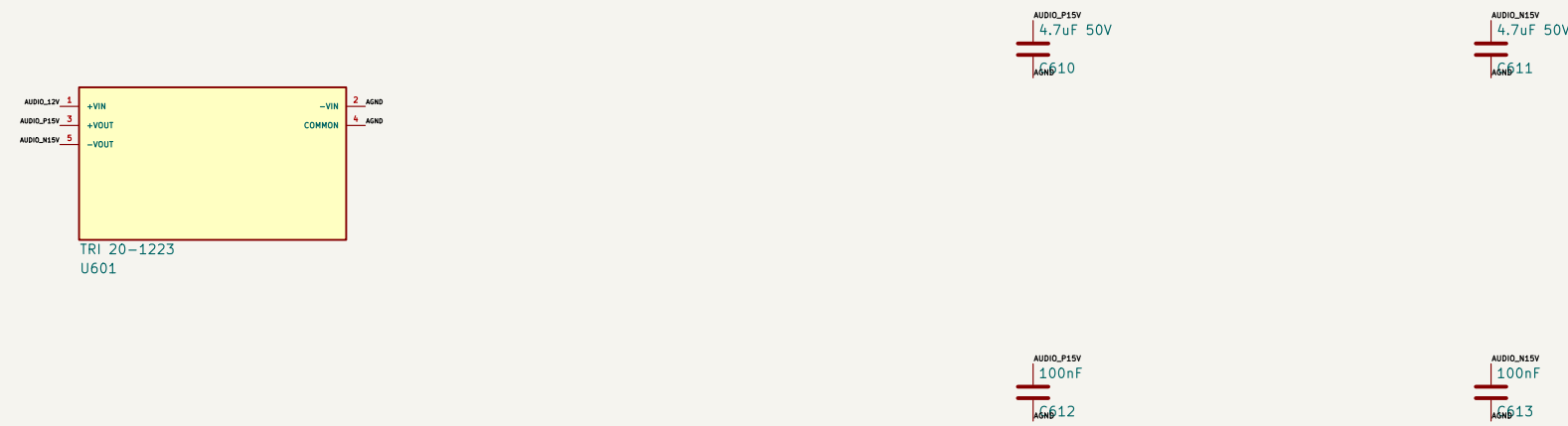
### 1. AUDIO\_12V ENTRY AND SINGLE GROUND STAR



R606 is a wide 2512 zero-ohm start; no fan, RF, Ethernet, or chassis current may share the AGND side.

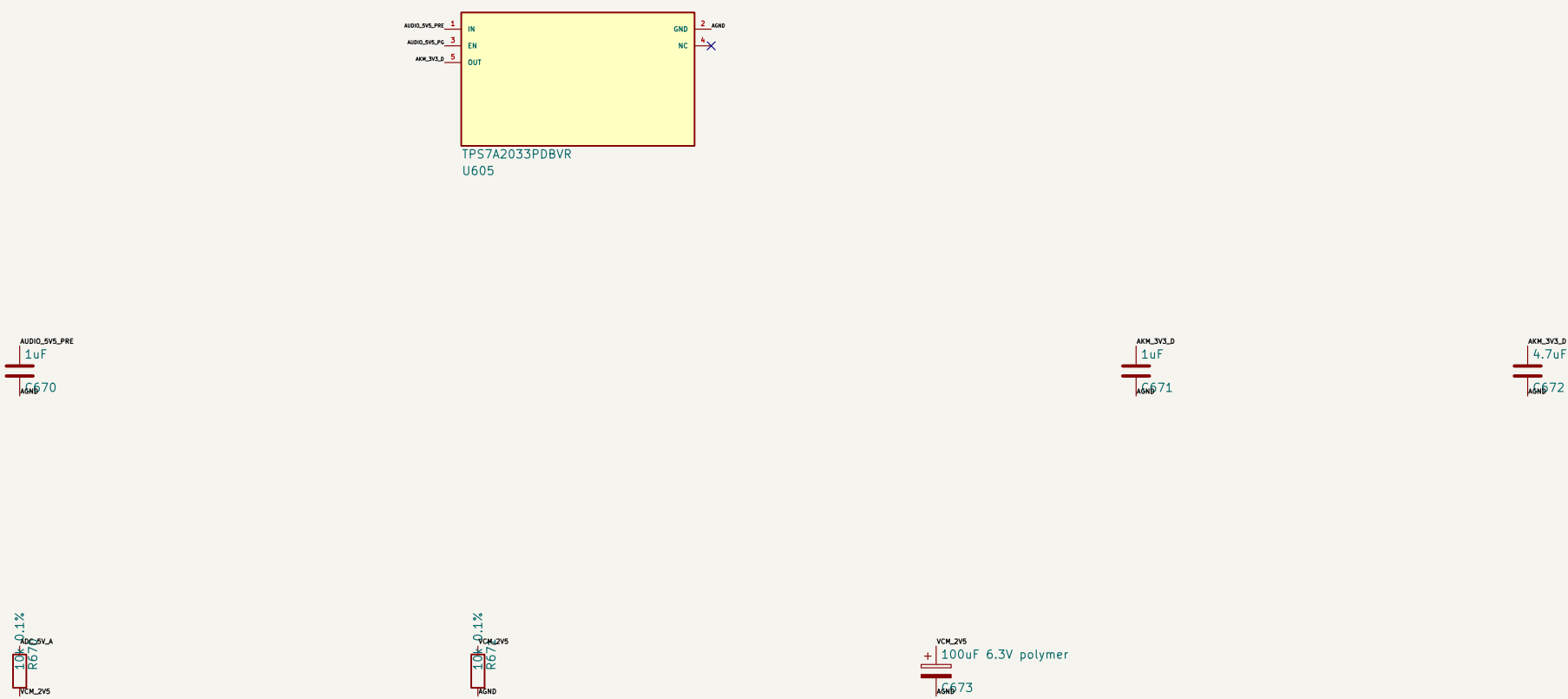
AUDIO\_12V source is separately fused on CRS-CARRIER; F606 protects this removable board and harness locally.

## 2. ISOLATED BIPOLAR LINE-STAGE RAILS



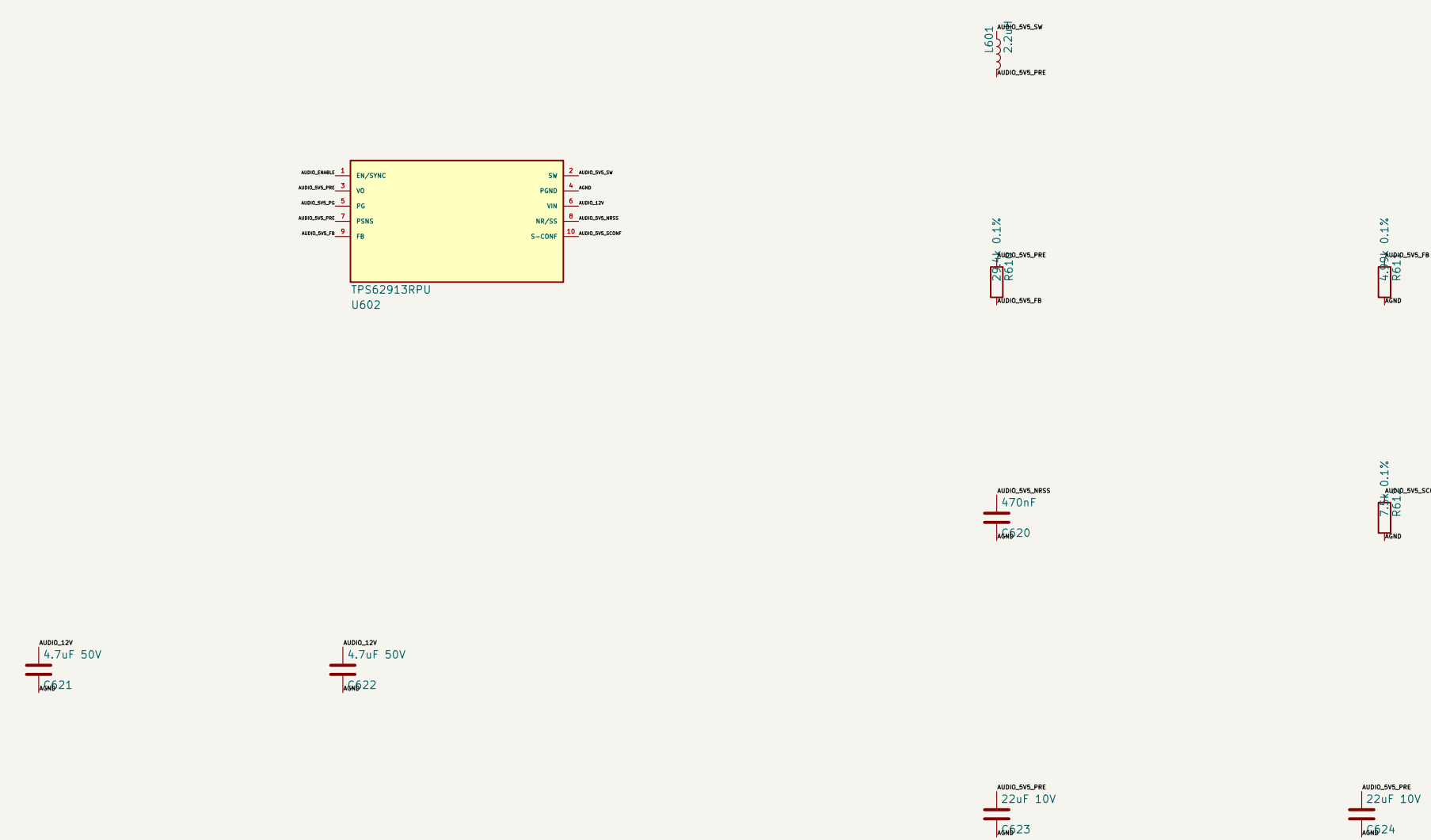
TRQ2: 9–18 V Input,  $\pm 15$  V at 670 mA per rail, 89% typical; place 4.7  $\mu$ F directly at each output pin.

## 5. AKM DIGITAL RAIL AND 2.5 V INPUT COMMON MODE



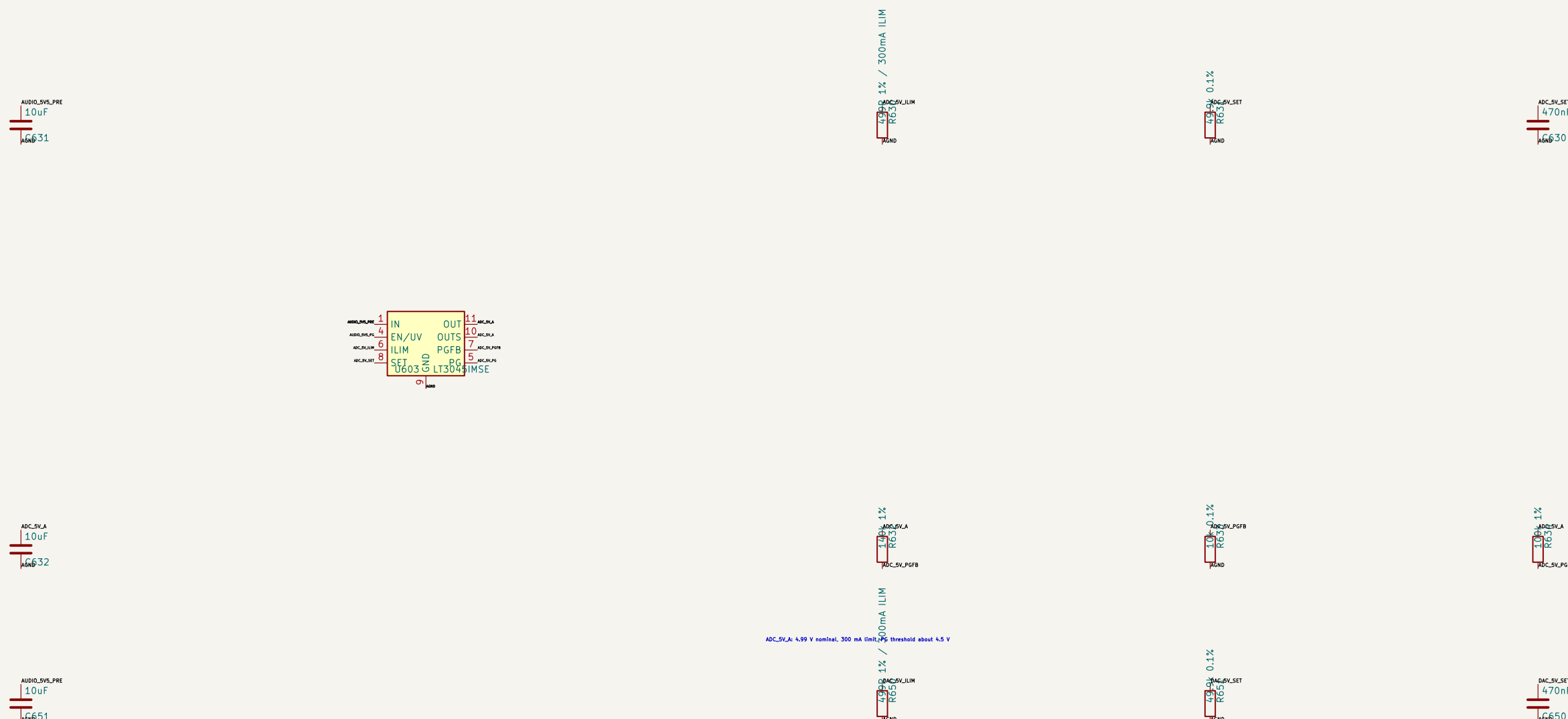
VCN2V5 is a quiet divider reference for the eight ADC input op-amps only; no external load is permitted.

### 3. QUIET 5.5 V PRE-RAIL



TPS62013: 2.2 MHz low-noise mode; 29.4k/4.99k feedback; 470 nF NP/SS; verify ripple below the AECM rail budget

#### 4. SEPARATE ADC AND DAC 5 V LT3045 RAILS



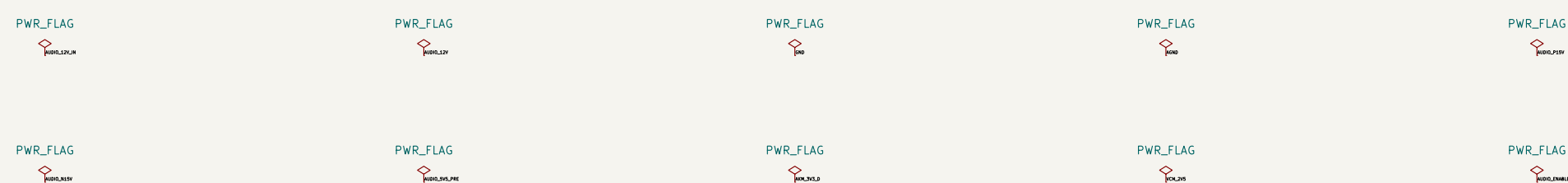
EDC<sub>3</sub>SV<sub>2</sub> at 4.99 V nominal, 300 mA limit,  $\tau_{\text{off}}$  threshold about 4.5

(EAL\_SVC): 4.99 V nominal, 330 mA limit, PG threshold about 4.5

## 6. ERC SOURCES AND SEQUENCING CONTRACT

Release under AUDIO\_ENABLE == YES/PC == (LTS445 PGs + AOM YES valid == clocks == resofs == SAC unmode == output relays

Any PG loss immediately de-energizes all output relays and reasserts OAC mode before software intervenes.



Power-ground connections: thermal rise, ripple, and conducted-emissions tests gate release  
Separate LT3045 5 V/300 mA rail feed ADC and DAC; TPS7A2033 feeds AKM digital I/O  
TRI 20-1223 creates +15 V; TPS62913 creates the quiet 5.5 V pre-rail  
AUDIO12V is the only carrier power input; local 3 A fuse and one GND-to-AGND star  
**ProComm**  
Sheet: /  
File: Audio-Power.kicad\_sch

**Title: ProComm AUDIO-8XB – Local Power, Sequencing, and Ground Star**

|                         |                  |         |
|-------------------------|------------------|---------|
| Size: A1                | Date: 2026-08-16 | Rev: A1 |
| KiCad E.D.A. 10.5.0-rc1 |                  | Ed: 1/1 |